

A REPORT
ON
MULTIMODAL PRODUCT CLASSIFICATION FOR E-COMMERCE
CATALOG

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MULTIMODAL PRODUCT CLASSIFICATION FOR E-COMMERCE
CATALOG**

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CERTIFICATE

This is to certify that the Project Work entitled **MULTIMODAL PRODUCT CLASSIFICATION FOR E-COMMERCE CATALOG** and submitted by **FEENAZ TASNEEM SAYED** ID No. **2023AA05354** in partial fulfillment of the requirements of AIMLCZG628T Dissertation, embodies the work done by him/her under my supervision.



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Abstract

This project aims to design and implement a **Multimodal framework** for automatic **product data classification** by integrating both textual and image features using Conventional models that rely on a single data modality often fail to capture the full semantic meaning of product information, leading to inaccurate classifications.

The primary objective of this project is to handle incomplete or ambiguous product descriptions in e-commerce platforms like companies client project that host millions of product listings that require accurate categorization to ensure efficient product discovery, recommendations, and inventory management. Traditional text-based classifiers struggle to handle incomplete or ambiguous product descriptions.

A multimodal approach offers significant advantages for e-commerce platforms, as it addresses the challenges of product categorization using merchant-provided images and labels, while also reducing duplication issues arising from the sale of both new and used products by professional and non-professional sellers. This problem presents several compelling research challenges due to the inherently noisy nature of product images and labels, the large scale of modern e-commerce catalogs, and the typically imbalanced distribution of product categories.

The proposed approach combines **Convolutional Neural Networks (CNNs)** for extracting visual features from product images and **Transformer-based models** for capturing contextual meaning from text data. These features are fused through various multimodal learning strategies to improve classification accuracy. The project also explores model evaluation under different fusion methods, performance comparison with unimodal baselines, and optimization for scalable e-commerce applications.



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LIST OF ABBREVIATIONS

AI	Artificial Intelligence
AIML	Artificial Intelligence and Machine Learning
BERT	Bidirectional Encoder Representations from Transformers
CNN	Convolutional Neural Network
CV	Computer Vision
DNN	Deep Neural Network
EDA	Exploratory Data Analysis
F1-score	Harmonic Mean of Precision and Recall
HTML	Hyper Text Markup Language
KNN	K-Nearest Neighbor
LR	Logistic Regression
ML	Machine Learning
MML	Multimodal Machine Learning
NLP	Natural Language Processing
RFC	Random Forest Classifier
RNN	Recurrent Neural Network
SVC	Support Vector Classifier
SMOTE	Synthetic Minority Over-sampling Technique
TF-IDF	Term Frequency–Inverse Document Frequency
VGG	Visual Geometry Group
API	Application Programming Interface
GPU	Graphics Processing Unit
CSV	Comma-Separated Values
ResNet	Residual Network
ImageNet	Large-Scale Image Database for Visual Recognition
prdtypecode	Product Type Code

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1. INTRODUCTION

1.1 General Overview

The rapid growth of e-commerce platforms has led to an exponential increase in the volume and diversity of online product listings. Efficient and accurate product classification plays a critical role in enabling effective search, recommendation systems, inventory management, and overall user experience. However, modern e-commerce catalogs often contain heterogeneous data in the form of textual descriptions and product images, which makes automated categorization a challenging task. In many real-world scenarios, product information provided by merchants is incomplete, inconsistent, or noisy, especially when listings are created by both professional and non-professional sellers.

Traditional product classification systems predominantly rely on unimodal approaches, using either textual information such as product titles and descriptions or visual features extracted from product images. While these approaches can be effective in limited cases, they often fail to capture the complete semantic representation of a product. Text-only models may struggle with ambiguous or poorly written descriptions, whereas image-only models may fail to distinguish visually similar products belonging to different categories. As a result, such systems suffer from reduced accuracy, limited generalization, and poor robustness in large-scale e-commerce environments.

To address these challenges, Multimodal Machine Learning (MML) has emerged as a powerful paradigm that integrates multiple data modalities to achieve better performance and reliability. Recent advances in deep learning, particularly Convolutional Neural Networks (CNNs) for image feature extraction and Transformer-based models such as BERT for textual understanding, have enabled the development of models that can jointly learn from visual and textual inputs. By fusing complementary information from both modalities, multimodal models can achieve a more comprehensive understanding of product attributes and improve classification outcomes.

This project, titled “**Multimodal Product Classification for E-Commerce Catalog**” focuses on designing and implementing an intelligent multimodal framework that combines text and image features to accurately classify products into predefined categories. The proposed system employs deep learning-based encoders for both modalities and integrates them through feature fusion techniques to enhance semantic representation. The work aims to improve classification accuracy, reduce duplication in product listings, and support automation in large-scale e-commerce platforms. Furthermore, the project provides insights into the impact of multimodal fusion on model performance and demonstrates the practical applicability of multimodal AI systems in real-world e-commerce scenarios.

1.2 Problem Statement

In large-scale e-commerce platforms, accurate product classification is a critical requirement for efficient catalog management, search, recommendation, and inventory control. However, existing product classification systems predominantly rely on **unimodal approaches** using either textual descriptions or product images independently. Such approaches often fail in real-world scenarios where product data is **noisy, incomplete, ambiguous, or inconsistent** especially when listings are created by diverse merchants, including both professional and non-professional sellers.

Text-based models struggle when product titles and descriptions are poorly written, missing, or misleading, while image-based models alone are insufficient to capture semantic and contextual information about products. This limitation results in **incorrect categorization, duplicate listings, reduced search relevance, and poor user experience**. Furthermore, e-commerce datasets typically exhibit **class imbalance and heterogeneous data distributions**, making reliable classification even more challenging.

Therefore, there is a need for an **intelligent multimodal classification framework** that can jointly leverage both **textual and visual information** to improve robustness and accuracy. The problem addressed in this project is to design and evaluate a **deep learning–based multimodal product classification system** that effectively integrates text and image features to achieve superior performance compared to unimodal approaches in real-world e-commerce environments.

2. LITERATURE SURVEY

Several researchers have investigated the application of artificial intelligence, machine learning, and deep learning techniques for automating product classification in large-scale e-commerce platforms.

Recent studies focus on **multimodal learning**, where textual and visual information are jointly analyzed to overcome the limitations of unimodal systems. Advances in **natural language processing**, particularly Transformer-based models such as BERT, have significantly improved the understanding of product titles and descriptions. Similarly, **convolutional neural networks** have demonstrated strong performance in extracting discriminative features from product images. Researchers have shown that integrating these modalities through feature fusion techniques leads to improved classification accuracy and robustness, especially in the presence of noisy, incomplete, or ambiguous merchant-generated data. This chapter reviews the most relevant research contributions related to multimodal product classification and e-commerce catalog automation.

Bader et al. proposed a multimodal product recognition framework that integrates visual features from product images and textual information extracted from product labels. The study highlights the limitations of unimodal systems, particularly in retail environments where products may be visually similar or labels may be partially occluded. Convolutional Neural Networks (CNNs) were used for image feature extraction, while text features were obtained using OCR-based and deep learning text processing techniques. These features were fused using deep neural architectures to improve classification robustness. Experimental results demonstrated that multimodal fusion significantly outperforms single-modality approaches in terms of precision, recall, and overall classification accuracy. This work strongly supports the motivation for combining text and image modalities in e-commerce product classification systems [1].

Ling et al. investigated the generalization of foundation models for e-commerce using high-quality multimodal instruction data.

The study emphasizes that textual captions often carry richer semantic information than images alone, especially for fine-grained product understanding. The authors proposed a multimodal learning framework that jointly trains vision–language models using curated image–text pairs. Their experiments showed improved performance across product classification and retrieval tasks when both modalities were used together. The research demonstrates that integrating textual descriptions with visual features enables better semantic alignment and robustness, reinforcing the importance of multimodal fusion techniques in large-scale e-commerce applications [2].

Several benchmark and challenge papers emphasize the importance of appropriate evaluation metrics for e-commerce classification tasks, particularly when class distributions are highly imbalanced. These studies recommend the use of weighted F1-score or macro F1-score instead of accuracy alone, as accuracy can be misleading in skewed datasets. Reporting per-class performance is also encouraged to identify underperforming categories. These findings are highly relevant to multimodal product classification, where category imbalance is common. Following these recommendations ensures a more reliable and fair evaluation of the proposed multimodal fusion model.

A. Shah, P. Goel, and V. C. Gandhi investigated image-based product classification using EfficientNetB5 with transfer learning. The authors demonstrated that EfficientNet architectures achieve high classification accuracy with fewer parameters compared to traditional CNNs such as VGG and ResNet. The model was trained on an e-commerce product dataset and showed improved performance through fine-tuning and data augmentation. Although the approach achieved strong results using visual features alone, the study acknowledges that image-only classification struggles with visually similar products. This limitation highlights the need for multimodal approaches, such as combining image models with text-based classifiers, as explored in the present project [3].

Pawłowski presented a machine learning-based framework for automated product classification in e-commerce platforms. The study analyzed both traditional machine learning algorithms and deep learning models using product metadata such as titles and descriptions.

The results showed that deep learning-based text classifiers outperform classical approaches in large-scale catalogs. However, the research primarily focused on textual data and did not incorporate visual information. The author emphasized challenges such as class imbalance and noisy labels, which remain critical issues in real-world e-commerce datasets. This work provides foundational insights and motivates the extension toward multimodal learning for improved classification accuracy [4].

Deep Residual Learning, a novel deep neural network architecture known as **ResNet**, designed to address the degradation problem encountered when training very deep convolutional neural networks by K. He, X. Zhang, S. Ren, and J. Sun. The authors propose residual learning blocks that use identity shortcut connections to allow gradients to flow directly through the network, enabling the successful training of networks with significantly increased depth. Experimental results on the ImageNet dataset demonstrate that residual networks achieve substantially higher accuracy compared to traditional CNN architectures while maintaining computational efficiency. The study establishes ResNet as a foundational model for image feature extraction and classification, making it highly effective for large-scale visual recognition tasks, including e-commerce product image classification [9].

The study highlights key challenges in multimodal learning, such as data heterogeneity, modality imbalance, synchronization issues, high computational cost, and lack of standardized evaluation benchmarks. It also identifies research gaps related to model interpretability, scalability, and robustness in real-world noisy environments. The paper concludes by outlining future research directions, emphasizing the need for efficient fusion techniques, better evaluation metrics, and domain-specific multimodal frameworks, which are highly relevant for large-scale e-commerce product classification systems.

3. PURPOSE OF THE WORK AND EXPECTED OUTCOME

3.1 Purpose of the Work and Expected Outcome

The purpose of this work is to design and develop an intelligent multimodal framework that can accurately classify e-commerce products by leveraging both textual and visual features. Conventional classification systems rely on only one modality—either product descriptions or images—which limits their ability to handle ambiguous, incomplete, or noisy data.

This project aims to overcome these limitations by integrating deep learning-based models for text and image understanding within a single architecture. The model will combine Convolutional Neural Networks (CNNs) for image feature extraction and Transformer-based or recurrent models for text representation. Through feature fusion techniques, the project seeks to enhance the semantic understanding of product data and improve the accuracy of automatic product categorization. It is useful technique for e-commerce companies as they have trouble categorizing products given images and labels from merchants and avoid duplication, especially when selling both new and used products from professional and non-professional merchants.

The expected outcomes include:

- A robust multimodal classification model that jointly learns from text and image features to produce accurate product category predictions.
- Improved classification accuracy and generalization performance compared to unimodal baselines.
- A reusable framework that can be adapted for other multimodal AI applications such as recommendation systems, visual search, and content tagging.
- Analytical insights into how multimodal fusion impacts performance and interpretability in real-world e-commerce environments

3.2 Existing Process and its Limitations

- In the current e-commerce landscape, most product classification systems rely primarily on **unimodal learning approaches**, where only **textual** or **visual** information is used to assign a product category.

- These systems typically follow one of the following workflows: **Text-Based Classification, Image-Based Classification, Rule-Based and Keyword Matching Systems.**

These methods have several limitations:

- Incomplete Representation of Product Information
- Low Accuracy in Ambiguous Cases
- Sensitivity to Noise and Missing Data
- Scalability and Maintenance Challenges
- Inability to Handle Multimodal Interactions

3.3 Project Work Methodology

The project follows these key steps:

- Input text (product title + description) and image.
- Text preprocessed and passed through Transformer encoder (BERT).
- Image resized and passed through CNN backbone (ResNet).
- Encoded vectors concatenated or fused in hybrid fusion layer.
- Final DNN classifier predicts product category.
- Evaluation metrics computed and performance analyzed.

3.4 Benefits Derivable from Work

- **Automation of Product Tagging and Categorization**
Reduces the need for manual classification and tagging of products, saving significant human effort and operational costs for e-commerce companies.
- **Improved Search and Recommendation Systems**
Accurate product categorization enhances the performance of search engines and recommendation systems, improving user experience and increasing conversion rates.

- **Enhanced Inventory Management**

Consistent product categorization supports efficient inventory tracking, demand forecasting and placement in digital catalog.

- **Cross-Platform Consistency**

The proposed system can be applied across multiple online marketplaces, to ensure the uniform classification and eliminating the duplication or inconsistency in product listings.

4. BROAD AREA OF WORK

Artificial Intelligence and Machine Learning - focusing on multimodal data integration and deep learning applications in e-commerce product classification.

Objectives

The objectives of my project are as follows:

- Understanding the context of the problem.
- Data pre-processing (cleaning, translation of text, categorizing).
- Data visualization.
- Feature Engineering (data transformation, lemmatization/stemming, tokenization, vectorization).
- Modeling (simple models, transfer models).
- Hyperparameter optimization.
- Unified product classification model (Fusion of text and image models).

5. SCOPE OF WORK

The scope of this work includes:

- Collecting and preprocessing product datasets containing **images and text attributes** (title, description, specifications).
- Understanding and maintaining data consistency, data cleaning, translation and exploration

are performed.

- Implementing **CNN-based visual feature extractors** and **Transformer or RNN-based text encoders**.
- Designing and training **fused model** that combines learned embeddings for text and image classification.
- Performing comparative evaluation using metrics such as **accuracy**, **F1-score** and **Confusion Matrix**.

Note: Since our company is not willing to share the data needed for the project, I will be working on the data which is available on the public domain.

6. EXPLORATORY DATA ANALYSIS

6.1 Understanding and Manipulation of Data

Textual Dataset:

The textual data used for training the model includes X_train and Y_train, containing a total of 84,916 records.

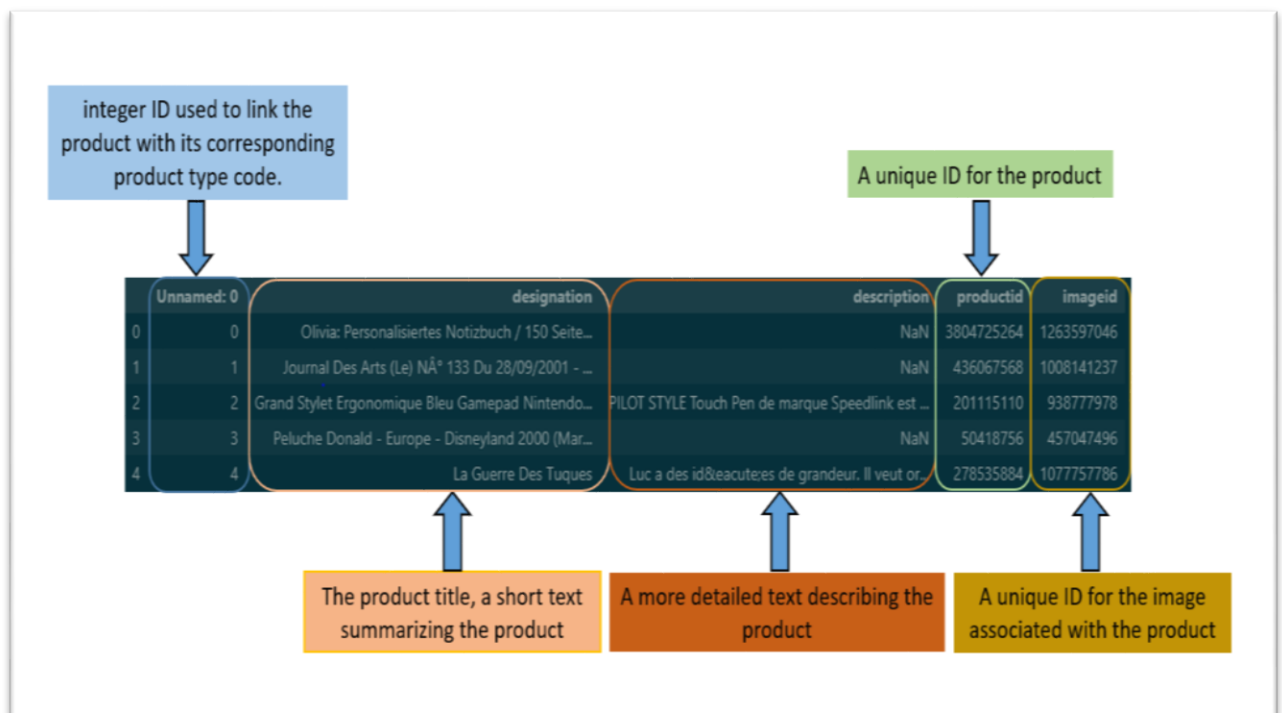


Figure 1. X_train dataset column information

X_train: This dataset includes textual information describing various products, such as the index ID (Unnamed: 0), designation, description, productid, and imageid.

Y_train: This dataset contains the product type code for the classification task, which is the target variable.

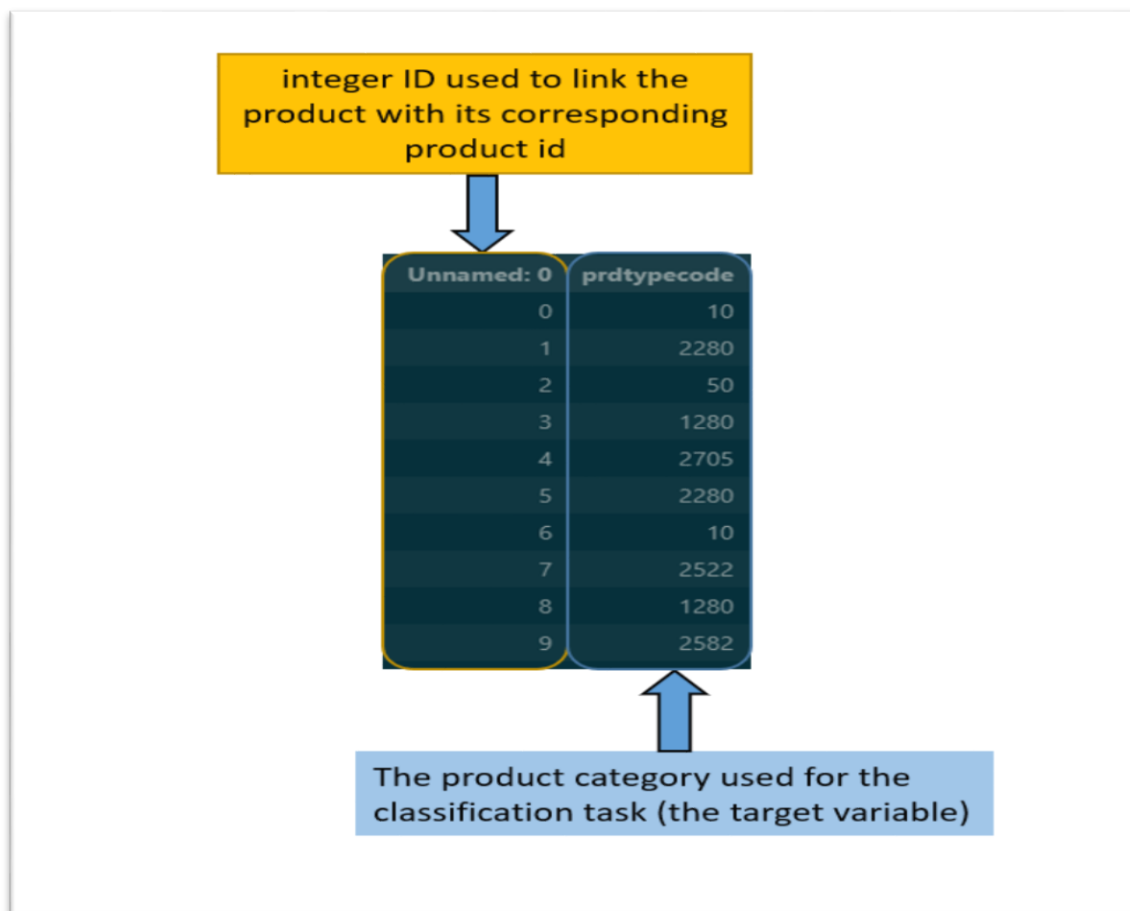


Figure 2. Y_train dataset column information

The X_train and Y_train datasets contain the following columns:

- Unnamed: 0 (index ID): An integer ID for the product. This ID is used to associate the product with its corresponding product type code.
- Designation: The product title, a short text summarizing the product.
- Description: A more detailed text describing the product. Not all merchants use this field, so some products may have NaN values in this column.

- Product ID: A unique ID for the product.
- Image ID: A unique ID for the image associated with the product.
- prdtypecode: The product category used for the classification task (the target variable).

Image Dataset:

The image dataset is provided as a compressed file containing 84,916 images. Each image is labelled with both a product ID and an image ID, which correspond to the entries in the **X_train** dataset.



Figure 3. Image information

6.2 Important Features for the Classification Tasks.

Textual Features

The textual data in the description and designation columns plays a crucial role in providing context for product classification. These features vary in length and complexity across different products, contributing significant information for determining the product's category.

1. Description: The **description** column contains important keywords, features, and detailed information about the product. This text is essential for understanding the product's characteristics and its associated category (**Product_type_code**).

2. Designation: The **designation** column consists of the product's title, which is a concise summary of the product. It provides critical clues for classification, such as the brand, model,

and specific **features** of the product, which are helpful for categorizing the product into the correct product type (**Product_type_code**).

Image Features

The image dataset consists of images named with corresponding imageid and productid, which are then used to retrieve the associated product_type_code for classification. The image features are integral for identifying the category of the product.

1. Image Resolution: The images have a resolution of 500x500 pixels.

2. Variability: The images vary in terms of sharpness, lighting, angles, and other visual factors. These variations provide valuable features such as colors, shapes, edges, and sharpness, all of which help the model learn and accurately classify the images into specific product categories.

Target Variable

The **target variable** for the classification task is the **product type code (prdtypecode)**, which is a categorical label used to categorize products into **27 distinct product categories** (e.g., pool accessories, books, kids' toys, etc.).

```
train = train.assign(product_category = train.prdtypecode)
train['product_category'] = train['product_category'].replace({
    10: "Used books",
    2280: "Used Newspapers and magazines",
    2483: "Books, comics, and magazines",
    2522: "Office supplies and stationery accessories",
    2785: "New books",
    40: "Video games, CDs, equipment, cables, new",
    50: "Gaming accessories",
    60: "Game consoles",
    2462: "Used video games",
    2985: "PC video games",
    1140: "Figurines, pop culture items",
    1180: "Trading cards",
    1160: "Board games and role-playing games",
    1280: "Children's toys",
    1260: "Children's board games",
    1281: "Model kits",
    1300: "Modeling",
    1325: "Outdoor games, air games, clothing",
    1560: "General furniture: beds, mattresses, sofas, chairs",
    2582: "Garden furniture: furniture and tools for the garden",
    1320: "Childcare, baby accessories",
    2567: "Pet supplies",
    2583: "Pool and accessories",
    2588: "Garden tools, outdoor technical equipment, house and pool accessories",
    1325: "Home linens, pillows, cushions",
    2560: "Decoration",
    1301: "Baby socks, small photos",
    1940: "Confectionery",
    2060: "Decoration",
    1920: "Household linen, pillows, cushions",
    2585: "Garden tools, Outdoor technical equipment for homes and swimming pools",
    1302: "Outdoor games, clothes",
    2220: "Pet store"
})
```

Figure 4. Product categories

The product category data for each product type code (prdtypecode) is collected from client's e-commerce dummy data.

Given the large number of product categories, this makes the classification task both challenging and interesting, requiring careful attention to both the textual and image data for effective model training.

6.3 Limitations Observed in the Dataset

Even though the provided dataset is enormous, following are some of the limitations and pre-processing tasks required on this dataset:

1. Imbalanced Data: Some product categories have more samples than others, which could lead to model bias where the model favors overrepresented categories. This may impact on the model's ability to generalize to rare or less-represented categories.

2. Image Quality: Some product images are of low-quality, poorly lit, or blank/solid color background or have an inconsistent background, which may limit the model's ability to extract meaningful features from them.

3. Missing or Inconsistent Data: Some products do not have associated images, and certain product descriptions are incomplete or missing, which could reduce the overall dataset size for training.

4. Multilingual Text Data: The product descriptions and names are primarily provided in French with some descriptions in other European languages. This poses challenges to train the NLP model with such a dataset and thus it requires language specific pre-processing.

5. Noisy Text Data: The product descriptions have variations in language and quality, such as spelling errors, missing information, and irrelevant terms (special characters and html tags), which could affect the model's ability to understand and classify the products accurately.

6.4 Data Preprocessing.

6.4.1 Text Data Preprocessing

6.4.1.1 Data Cleaning.

Checked for duplicates and missing (NaN) values in the dataset.

6.4.1.2 Handling Missing Data

35.09% missing data was observed in the product description column. Given the dataset's imbalance and the importance of this feature for classification, a new column 'title_description', was generated by concatenating the two columns product designation and description.

6.4.1.3 Text Pre-processing

- **Translation:** As the text data is multilingual (French/German and English), GoogleTrans API was used to translate the content in the 'title_description' column to English language and is stored in the column 'description_english'.
- **Data Cleaning:** Before translation the text data was cleaned by applying Latin encoding to restore special characters, removing HTML tags, special characters, and punctuation using regular expressions, and converting the text to lowercase.

6.4.1.4 Pre- processing steps for Exploratory Data Analysis

- **Tokenization:** Performed using a simple tokenizer and converted all tokens to lowercase.
- **Stop Words Removal:** Common stop words (e.g., "and", "the", "a", etc.) were removed, as they do not contribute significantly to model learning.
- **WordCloud:** Created WordCloud to visualize the top 200 most frequent words in the product title/description.

6.1.4.5 Pre-processing steps for Text Classification Model

- Performed tokenization on the translated text, converted to lowercase, and applied stemming and lemmatization for reducing words to their root forms.
- Applied vectorization using CountVectorizer or TF-IDF (as these methods do not require normalization or standardization).

6.5 Image Data Pre-processing

6.5.1 Image Filename Information

Extraction Each image filename contains two key pieces of information: image ID and product ID (e.g., image_1263597046_product_3804725264).

6.5.2 Linking Images to Product Categories

- The image ID and product ID were extracted from the filenames to link each image to its corresponding product category (or prdtypecode).
- The following steps were carried out to associate the images with the correct product category
 - Extracted the image path and filename for each image from the image folder.
 - From each filename, extracted the image ID and product ID, storing them in the new 'image_ID' and 'product_ID' columns.

6.5.3 Handling Blank Images

Identified and removed 7 blank/solid images based on pixel value comparison.

6.5.4 Image Feature Extraction.

Extracted features such as image size, pixel resolution, sharpness (calculated using Laplacian variance), and storing them in respective new columns.

6.7 Visualizations and Statistics

6.7.1 Relationship between variables

The relationships between the text variables and the target classes were thoroughly analyzed, revealing significant correlations between specific terms and product categories. Advanced text embedding techniques were employed to effectively capture these relationships. Additionally, image data was incorporated, providing valuable contextual features that contributed to improving classification accuracy.

6.7.2 Textual Data Exploration

6.7.2.1 Imbalance in product category or prdtypecode

The dataset is categorized into 27 product type categories, and the number of products in each category is depicted in the figure below.

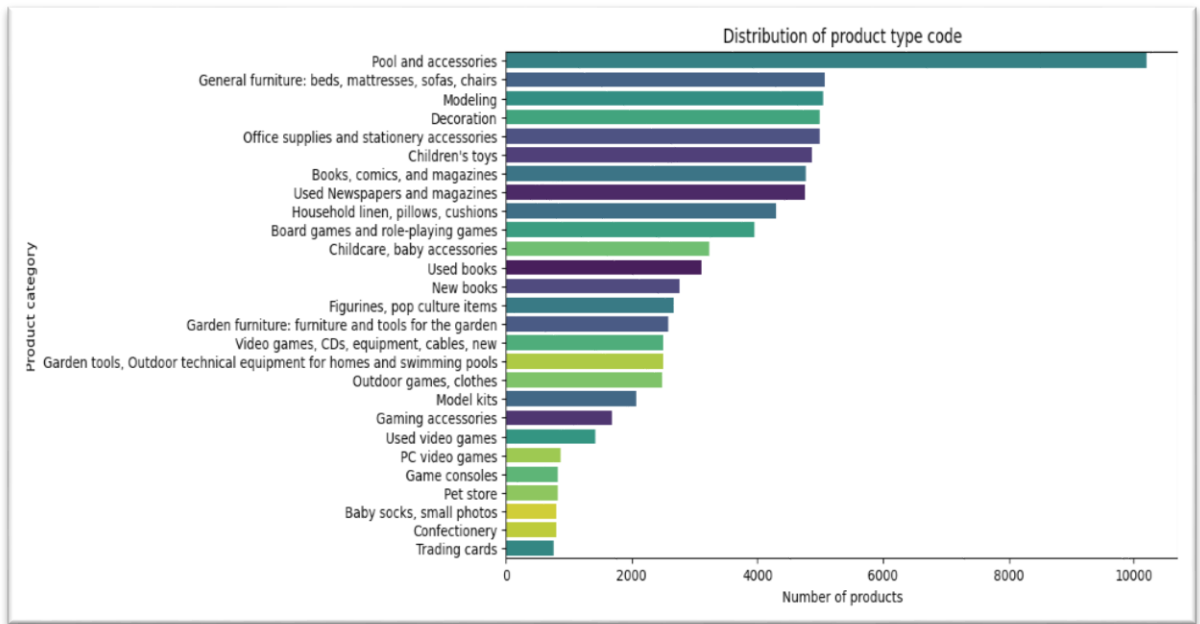


Figure 7. Distribution of product designation length in each category

From the figure, the following observations can be made:

- The **"Pool and Accessories"** category has the highest number of products, with a total of 10,209 items.

- The "**Trading Cards**" category has the least number of products, with only 764 items, among all the other categories.

6.7.2.2 Distribution of Product Title and Description Text

The box plots below illustrate the distribution of product title (designation) and description text lengths across different product categories in the dataset.

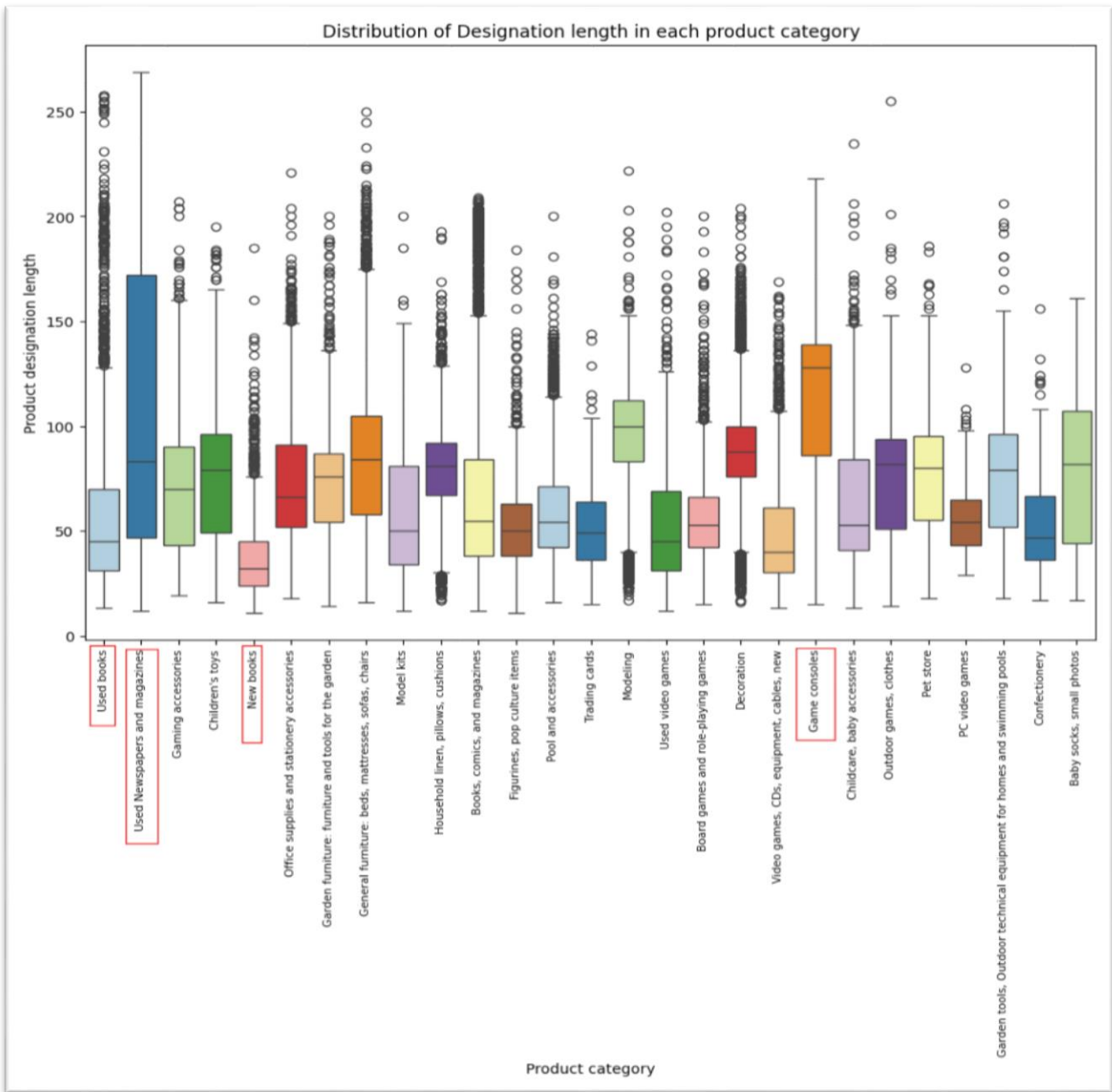


Figure 8. Distribution of product designation length in each category

From the above figure, the following observations have been made:

- The "**Games Consoles**" category has the longest designation length (high median value) with the fewest outliers (i.e., only 1).
- The "**Used Newspaper and Magazines**" category exhibits a wide range of designation lengths.
- The "**Used Books**" category has a smaller median designation length but also shows larger variance (outliers) up to 250 characters.
- The "**New Books**" category has the shortest designation length on average.
- Product category "**New Books**" has minimum characters in designation.

From the below figure, the following observations have been made:

- The "**PC Video Games**" category has the longest description length (high median value), with an average of 2,500 characters.
- The "**Used Newspaper and Magazines**" category again shows a wide range of description lengths.
- Most products in the following categories lack descriptions (median value is 0): "**Used Books, Used Newspapers and Magazines, Books, Comics and Magazines, Used Video Games, Board Games, CDs, Equipment, Cables, New**".
- The "**Children's Toys, Modeling, Decoration**" categories contain many outliers in terms of description length.
- The longest description was found in a product from the "**General Furniture, Beds, Mattresses, Sofas, Chairs**" category.

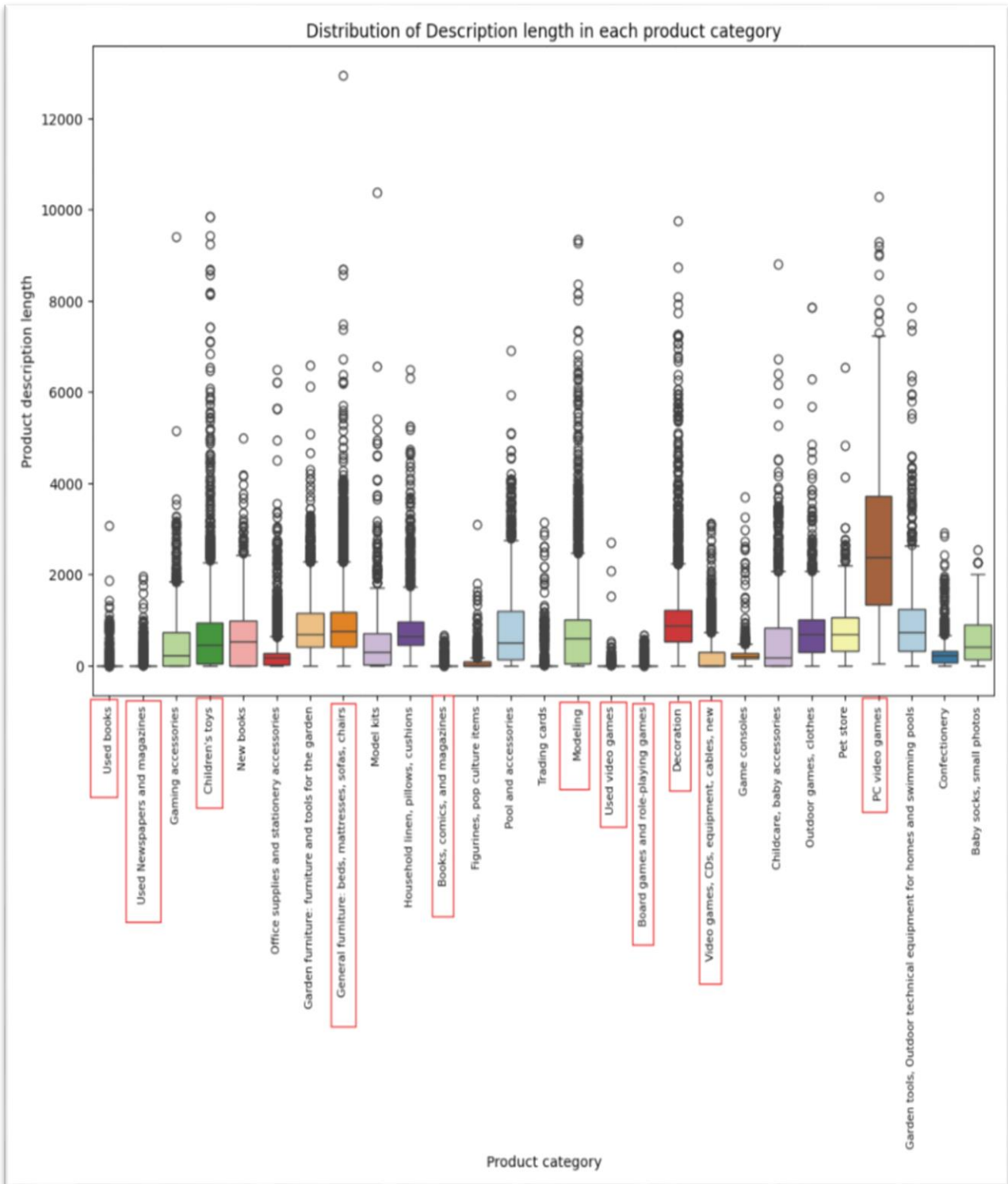


Figure 9. Distribution of product description length in each category

6.7.2.3 Distribution of description in product categories (Proportion of products with and without description)

As observed in 9, some products do not have descriptions, as indicated by a median value of 0. To further illustrate this, a bar plot is presented to show the proportion of products with and without descriptions across different product categories.

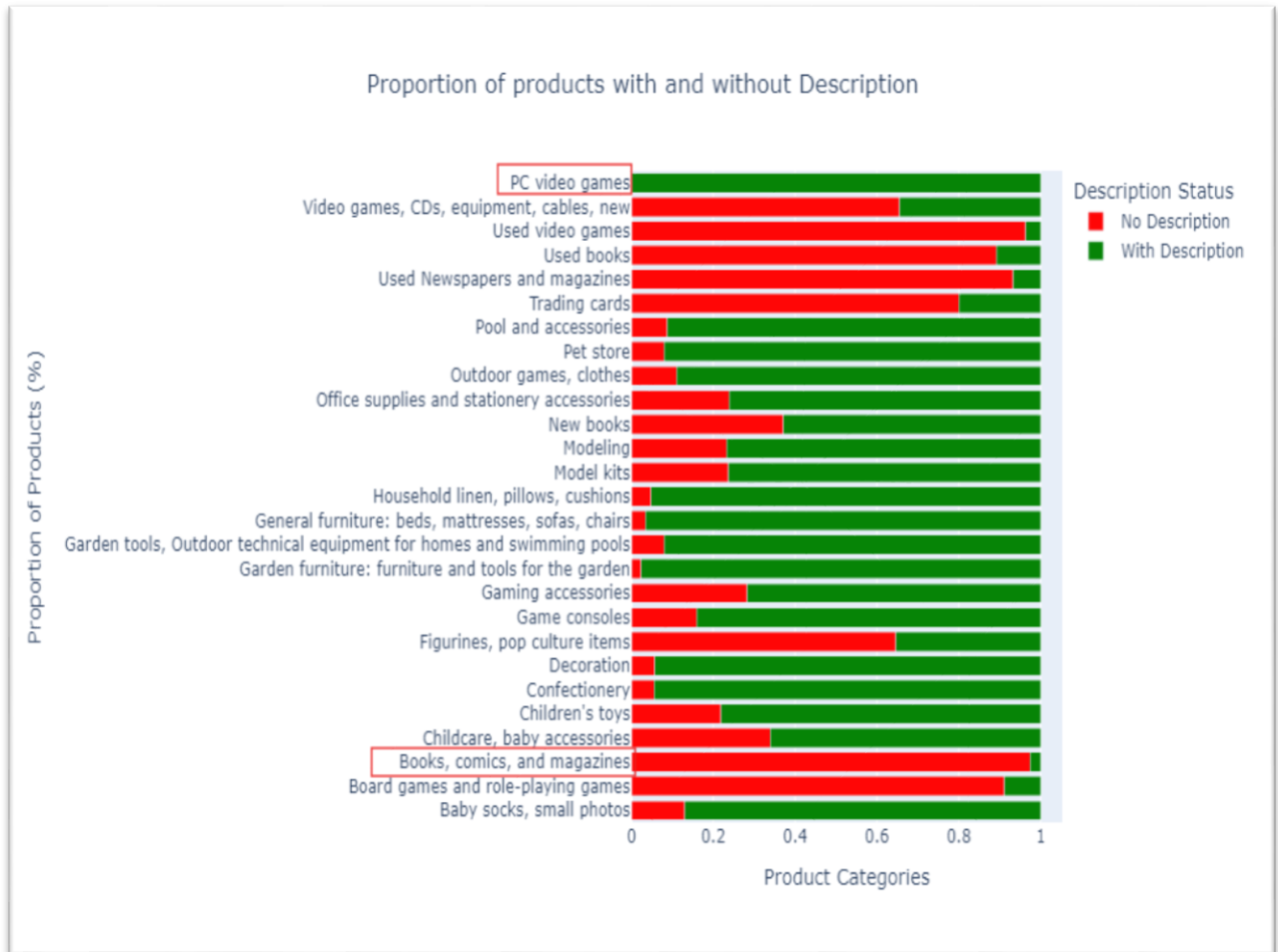


Figure 10. Proportion of products with and without description

The plot clearly shows that certain product categories are missing product descriptions. The **'Books, comics and magazines'** category have the highest proportion of products without descriptions, while **'PC Video Games'** contains only products with descriptions.

6.7.2.4 WorldCloud of frequent words in product title/description

After pre-processing the textual data, which involved the removal of special characters, HTML tags, and punctuation (using the regular expressions library), conversion to lowercase, removal of common stop words, stemming, and translation to English, a WordCloud was generated to visualize the most frequent words in the translated product titles for all products.



Figure 11. WordCloud of frequent words in product title/description

The WordCloud above highlights the 200 most frequent words in the product titles/descriptions. Notably, the words ‘package,’ ‘include,’ ‘stainless steel,’ and ‘brand’ appear frequently and are among the most repeated words across all product titles and descriptions.

6.7.3 Image Data Exploration

6.7.3.1 Removal of 7 blank Image from the Image Dataset

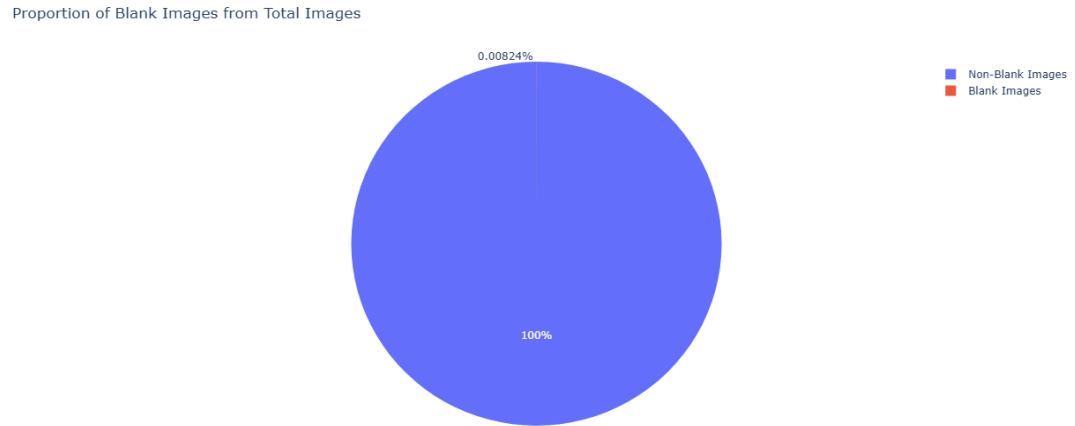


Figure 12. Proportion of blank images from total images

During the exploration of the image dataset, it was discovered that some product images were either blank or consisted of a solid background. To identify and omit these blank or solid images, each image was categorized as "blank" or "not blank" by comparing the pixel values. A bar plot was generated to visualize the distribution of blank images across all product categories.

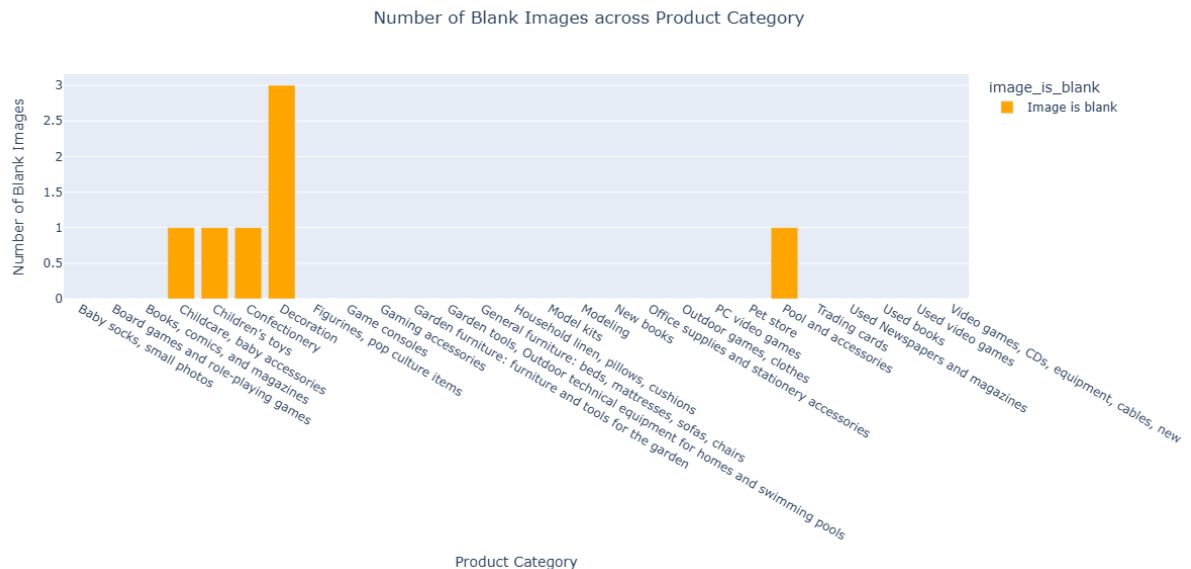


Figure 13. Number of blank images across product categories

6.7.3.2 Distribution of product categories Image with and without text

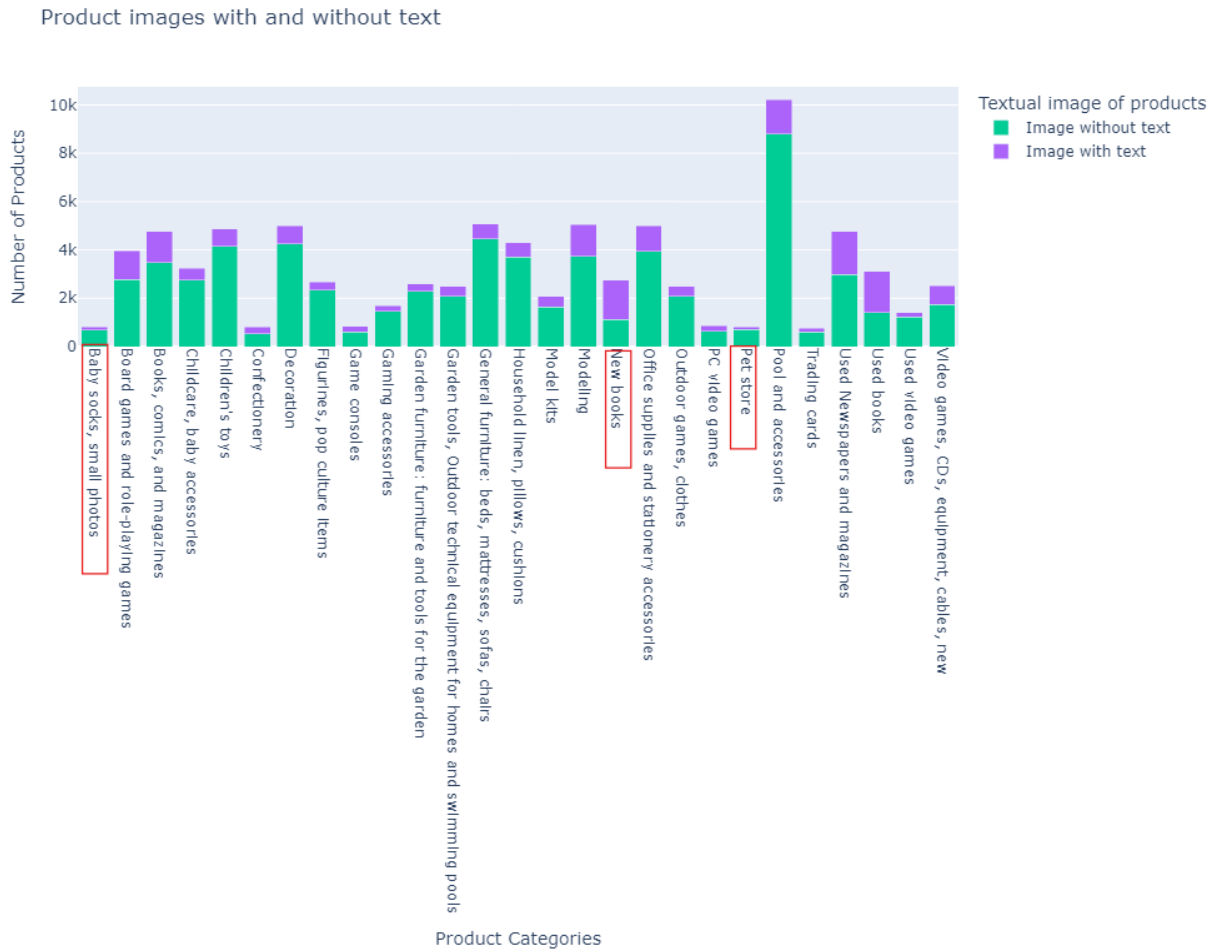


Figure 14. Product images with and without text

From the above figure, the following observations have been made:

- As expected, category '**New books**' have more text on images.
- **Pet Store/ Baby socks, small photos** category have least number of texts on their product images.

6.7.3.3 Proportion of image sharpness for each category

Since image quality impacts the performance of learning models, a study was conducted to assess the quality of the images. The quality of each image was measured by calculating the Laplacian variance, which measures the rate of change in pixel intensity. Based on the Laplacian variance value, the images were categorized into the following categories: sharp (≥ 200), moderately sharp (199 - 100), slightly blurry (99 - 50), and blurry (< 50). The proportion of images in each sharpness category is illustrated in the pie chart below.

Proportion of Images in Each Sharpness Category

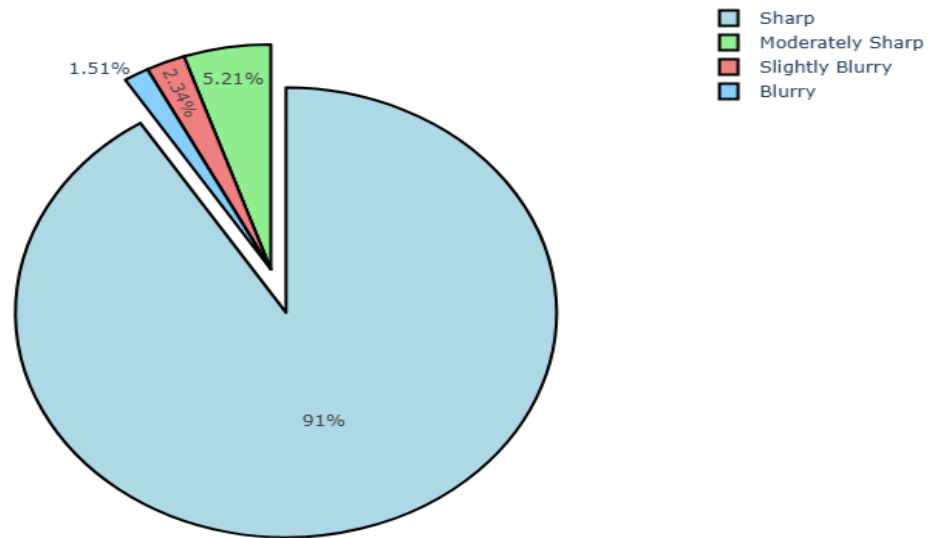


Figure 15. Proportion of Images by sharpness category.

It can be observed from the pie chart below:

- The majority (91%) exhibited sharp, high-quality characteristics.
- Among the images, a small proportion (1.51%, or 1,280 images) were classified as blurry and 5.21% were moderately sharp, and 2.34% were slightly blurry.

6.7.3.4 Distribution of image sharpness score in each product category

It can be observed from the above Boxplot:

- The product category ‘**Garden Tools, Outdoor Technical Equipment for Homes and Swimming Pools**’ has the highest sharpness score (i.e., 60,000).
- The ‘**Pool and Accessories**’ category has the lowest image sharpness.
- The ‘**PC Video Games**’ category exhibits the highest median value for image sharpness scores, indicating that a larger proportion of images in this category are of higher quality.

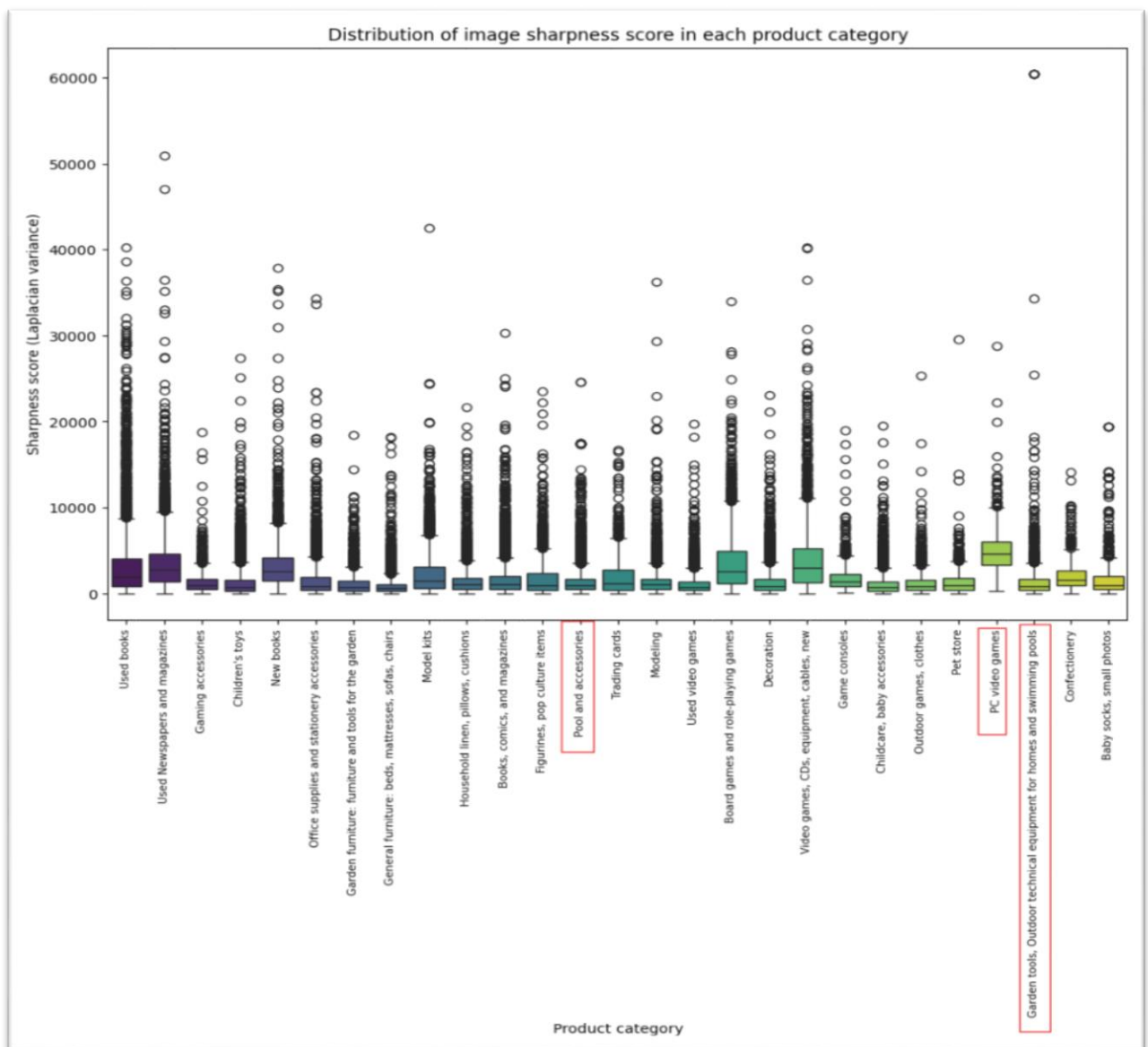


Figure 16. Distribution of Image sharpness across all product categories

7. MODEL STRATEGY AND TESTING

7.1 Strategy for Model Testing and Selection

The Project employed a structured strategy for model testing. Initially, the traditional machine learning models were tested and evaluated their performance. Subsequently, advanced transfer learning models were applied to improve classification accuracy. Below is an overview of how different models were tested, optimized, and evaluated.

7.2 Features and Target variables

For Text classification model, the column ‘description_english’ is used as the feature and the column ‘prdtypecode’ is used as the target variable.

For image classification model, the ‘image_path’ which directs to the image file was used as feature and ‘prdtypecode’ as target variable.

7.3 Handling Class Imbalance

For the **textual dataset**, after pre-processing, the class imbalance is addressed by applying the **SMOTE** (Synthetic Minority Over-Sampling Technique) method for machine learning models, and the class_weight method is used for transfer learning models.

For the **image dataset**, the imbalance is tackled by generating three augmented images for each image in the minority classes. After splitting the dataset into training, validation, and test sets, data augmentation is applied to the training set, irrespective of class imbalance, to enhance classification performance.

7.4 Classification of the Problem

The project relates to **multimodal classification**, combining both text classification and image classification tasks.

Example for a Multimodal Classification

- **Text Classification**

DistilBERT can be used for classifying textual data into predefined categories.

- **Image Classification**

VGG16 can be employed to categorize images into different classes.

7.5 Metric used to compare the Models

The following metrics are used to evaluate and compare model performance:

- **Validation Accuracy:** Measures the overall correctness of the model on the validation set.
- **Validation Loss:** Assesses how well the model minimizes errors during training.
- **F1-Score:** A balanced measure of precision and recall, especially useful for imbalanced datasets.

7.6 Machine Learning Model Results for Text Classification

For Text Classification, several pre-processing techniques were applied, including **stemming** and **lemmatization**. Both datasets were split into a training set (80%) and a test set (20%) using the `train_test_split` method from Scikit-learn. To address the imbalance in the dataset, the **SMOTE** resampling method was used for simple machine learning models. The results from the stemmed dataset showed lower accuracy and F1-score compared to the lemmatized dataset. Various trials with different Machine learning models were carried out and the results can be referred in Appendix table(a).

As a result, the lemmatized dataset was selected for further experimentation. Two vectorization methods, **CountVectorizer** and **TF-IDF**, were used with various machine learning models.

Result found for the **Lemmatized** dataset with **CountVectorizer** with the models and the respected output with accuracy and F1-score is as shown below.

	Model	Accuracy	F1 Score
0	LogisticRegression	0.659974	0.668793
2	RandomForestClassifier	0.654675	0.660498
1	SVC	0.626178	0.643189
5	MultinomialNB	0.614461	0.620009
4	GradientBoostingClassifier	0.587965	0.613926
3	KNeighborsClassifier	0.529027	0.547900

Result found for the **Stemmed** dataset with **TF-IDF** with the models and the respected output with accuracy and F1-score is as shown below.

	Model	Accuracy	F1 Score
1	SVC	0.704663	0.710765
0	LogisticRegression	0.685999	0.691595
2	RandomForestClassifier	0.660210	0.666557
5	MultinomialNB	0.661682	0.664834
4	GradientBoostingClassifier	0.616463	0.644822
3	KNeighborsClassifier	0.508596	0.535548

Result found for the **Lemmatized** dataset with **TF-IDF** with the models and the respected output with accuracy and F1-score is as shown below.

	Model	Accuracy	F1 Score
1	SVC	0.709550	0.713525
0	LogisticRegression	0.690709	0.694255
2	RandomForestClassifier	0.662800	0.669027
5	MultinomialNB	0.663625	0.664221
4	GradientBoostingClassifier	0.617876	0.645200
3	KNeighborsClassifier	0.492522	0.519858

Among the models tested with **TF-IDF vectorization**, **SVC**, **Logistic Regression**, and **Random Forest Classifier delivered the best accuracy and F1-scores**, outperforming others like KNN, Multinomial Naive Bayes, XGBoost, and GaussianNB.

Table 1. Machine learning models and results for text classification

Tokenization/Vectorization	Models	Validation accuracy	F1-score
Simple Tokenizer, Countvectorizer	SVC	0.626178	0.643189
Simple Tokenizer, Countvectorizer	Logistic Regression	0.659974	0.668793
Simple Tokenizer, Countvectorizer	RFC	0.654675	0.660498
Simple Tokenizer, Countvectorizer	Multinomial NB	0.614461	0.620009
Simple Tokenizer, Countvectorizer	XGBoosting	0.587965	0.613926
Simple Tokenizer, Countvectorizer	KNN	0.529027	0.547900
Simple Tokenizer, TF-IDF Vectorizer (features= 5000)	SVC	0.709550	0.713525
Simple Tokenizer, TF-IDF Vectorizer (features= 5000)	Logistic Regression (max_iter=200)	0.690709	0.694255
Simple Tokenizer, TF-IDF Vectorizer (features= 5000)	RFC (n_estimates=50)	0.662800	0.669027
Simple Tokenizer, TF-IDF Vectorizer (features= 5000)	KNN	0.5457	0.5682
Simple Tokenizer, TF-IDF Vectorizer (features= 5000)	Gaussian NB	0.5253	0.5299

* SVC - Support Vector Classifier, RFC – Random Forest Classifier, KNN – K-Nearest Neighbor

Advanced models such as deep learning models were tested for text classification. For text classification, BERT variant DistilBERT is tested to their superior performance in natural language processing tasks. Here 70% or 80% of datasets are used for training, and the remaining was split into half, where the first half is used for validation and second half is used for testing the model. DistilBERT yielded the best performance with an accuracy of 0.83 as shown in Table 2.

7.7 Advanced NLP Models

Various BERT variants for text classification were tested, among which DistilBERT, ALBERT, and RoBERTa were the primary models tested for text classification. The models were trained on 80% of dataset for training and 20% for validation.

Below are the results for the BERT variants tested for text classification.

Table 2. NLP BERT result for text classification

Tokenization/ Vectorization	Models	Validation Accuracy
DistilBERT	DistilBERT-base-uncased (learning rate= 3e-5) (Epoch 4)	0.8414
ALBERT	Albert (learning rate= 1e-5) (Epoch 4)	0.7998
RoBERTa	Roberta (learning rate=1e-5) (Epoch 4)	0.8166

DistilBERT yielded the best performance with a validation accuracy of 84%, as shown in the table above.

7.8 Optimization Results for DistilBERT

Various hyperparameters were experimented with for DistilBERT, including learning rates, batch sizes, and train-validation-test splits. Here 70% or 80% of datasets are utilized for training, and the remaining was split into half, where the first half is used for validation and second half is used for testing the model.

Table 3. Optimization results for DistilBERT-base-uncased model

Tokenizer	Models	Test Accuracy	F1-Score
DistilBERT	DistilBERT-base-uncased (learning rate= 4e-5), Epoch =6, batch =32 (70, 15, 15 split)	0.8321	0.8174
DistilBERT	DistilBERT-base-uncased (learning rate= 1e-6), Epoch =10, batch =32 (80, 10, 10 split)	0.7440	0.6817
DistilBERT	DistilBERT-base-uncased (learning rate= 4e-5), Epoch =6, batch =16 (70, 15, 15 split)	0.8350	0.8209
DistilBERT	DistilBERT-base-uncased (learning rate= 3e-5), Epoch =6, batch =16 (80, 10, 10 split)	0.8400	0.8236
DistilBERT	DistilBERT-base-uncased (learning rate= 3e-5), Epoch =6, batch =32 (80, 10, 10 split)	0.8387	0.8236
DistilBERT	DistilBERT-base-uncased (learning rate= 4e-5), Epoch =10, batch =32 (80, 10, 10 split)	0.8434	0.8294
DistilBERT	DistilBERT-base-uncased (learning rate= 5e-5), Epoch =10, batch =32 (80, 10, 10 split)	0.8482	0.8382

Among the different configurations, the best performance was achieved with an F1 score of 83.82, using a data split of 80% train, 10% validation, and 10% test, learning rate of 5e-5, a batch size of 32, and epoch 10.

7.9 Advanced Models for Image Classification

In the initial approach, a custom CNN model was developed, but its performance was suboptimal with a very low validation accuracy (< 0.20). After 10 epochs, the F1-score obtained is 0.18 which is less than 0.2 and the computational time was around 13 hours to run the 10 epochs.

```
print("Weighted F1 Score:", f)
```

Weighted F1 Score: 0.1884960193

Also, CNN models like ResNet50 and VGG16 were trained. These deep learning models were chosen because they are pre-trained on large datasets like ImageNet, which contains millions of images and thousands of classes, making them more efficient and accurate for classification tasks.

7.10 VGG16 performed better for Image Classification

For Image Classification, 80% of dataset for training and the remaining was split into half, where the first half is used for validation and second half is used for testing the model. The initial approach used a custom CNN model, but its performance was suboptimal with a very low validation accuracy (< 0.20). As a result, pre-trained transfer learning models such as ResNet50 and VGG16 were employed, both with their original and reduced base layers. To address class imbalance in the dataset, ImageDataGenerator was used to generate augmented data for the minority classes in the training set, following the train_test_split. Among the tested models, the results from ResNet50 and VGG16 showed varying levels of accuracy, with VGG16 outperforming ResNet50 in terms of accuracy, F1 score and validation loss under specific conditions. The validation accuracy and loss obtained from the ResNet50 and VGG16 models with different base layers and learning rates is shown in the table below.

Table 4. Image classification models and results

Model	Learning Rate	Validation Accuracy	Validation Loss
Resnet50 (Epoch 10)	1e-4	0.2668	2.5473
Resnet50(base_layer:-10) (Epoch 10)	2.50E-05	0.414	2.0206
Resnet50(base_layer:-10) (Epoch 10)	3.13E-06	0.4316	1.9658
Resnet50(base_layer:-10) (Epoch 10)	1.95E-07	0.4345	1.9563

VGG16 (Epoch 25)	1e-4	0.5500	1.5377
------------------	------	--------	--------

Among the ResNet50 and VGG16 models, the best accuracy 0.5500 and F1 score 0.5374 were achieved using the VGG16 model with a learning rate of 1e-4, a batch size of 16, and 25 epochs.

8. INTERPRETATION OF RESULTS

Text Model: DistilBERT-base-uncased (learning rate = 5e-5, Epoch = 10, batch_size = 32, split = 80/10/10)

Image Model: VGG16 (learning rate = 1e-4, Epoch = 25, batch_size = 16, split = 80/10/10)

8.1 DistilBERT-base Model for Text Classification

Architecture: DistilBERT is a smaller, faster version of BERT (Bidirectional Encoder Representations from Transformers), it is designed to reduce the size and computational cost while

maintaining performance.

Use Case: Primarily used for Natural Language Processing (NLP) tasks such as text classification,

sentiment analysis and named entity recognition.

Advantages:

- **Faster** and more **memory-efficient** than BERT.
- Retains **over 95% of BERT's performance** despite being smaller.
- Ideal for NLP tasks where speed and efficiency are critical.

Confusion Matrix

The analysis of misclassified records provides key insights into the model's behavior. The confusion matrix obtained from the test dataset evaluation shown in Figure 10.

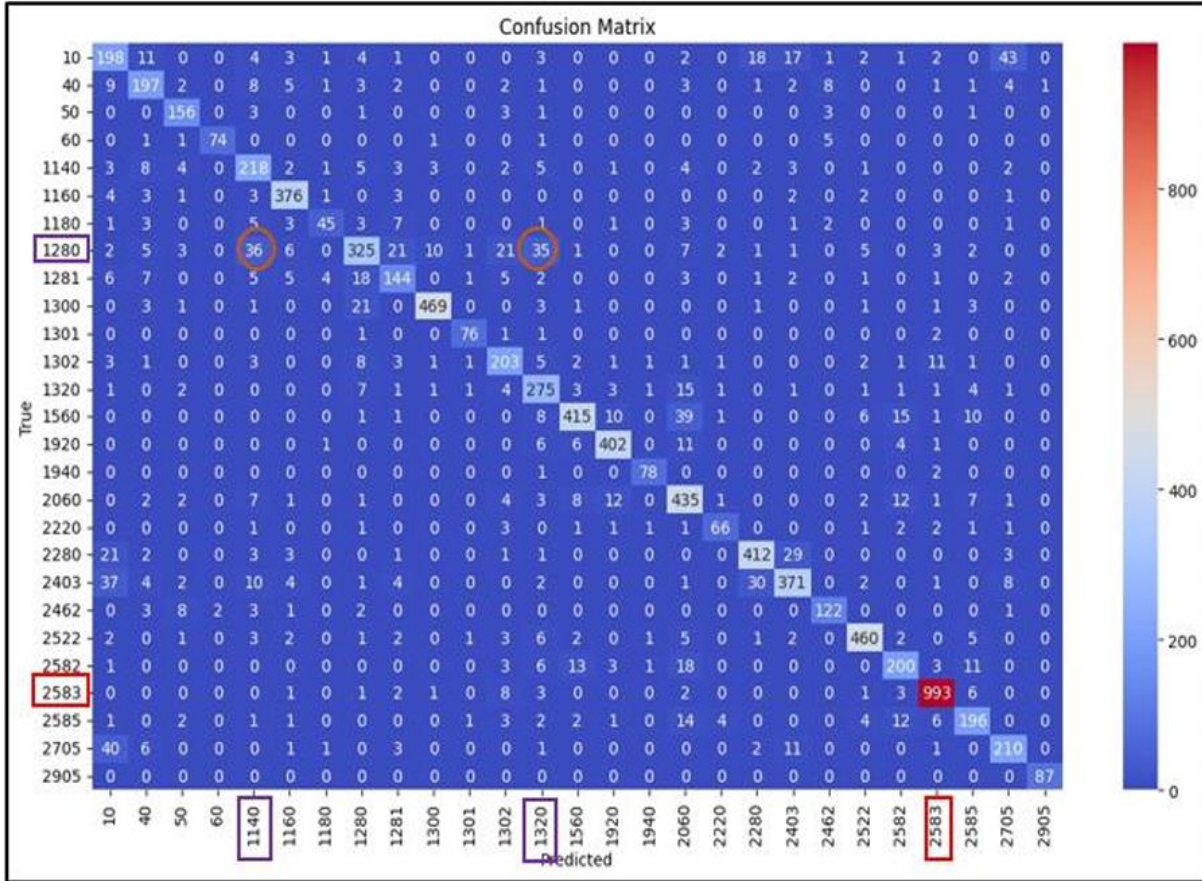


Figure 17. Confusion matrix for text classification model

Classification Statistics

The highest number of correctly classified products was observed in the "**Pool and Accessories**" category (prdtypecode: 2583). This category represents the majority class, with a total of 10,209 records in the dataset.

Misclassification Statistics

Analyzing misclassified records provides valuable insights into the model's performance and limitations. Among the misclassified records, the **Children's Toys** category (prdtypecode: 1280) has the highest number of errors, with **162 instances**. Most of these are misclassified into

the **Childcare and Baby Accessories** category (prdtypecode: 1320) and **Figurines** (prdtypecode: 1140). This misclassification is understandable since both categories deal with children's products.

Interestingly, the **Children's Toys** category is not a minor class it accounts for a significant **4,870 records** in the dataset.

From the confusion matrix, another notable observation is that the highest number of products misclassified into a single category belongs to the **General Furniture** category (prdtypecode: 1560). Specifically, **39 products** were incorrectly classified as part of the **Decoration** category (prdtypecode: 2060).

8.2 VGG16 Model for Image Classification

Architecture: VGG16 is a deep convolutional neural network (CNN) with 16 layers, consisting of convolutional layers followed by fully connected layers.

Use Case: Commonly used for image classification tasks, especially with large datasets like ImageNet.

Advantages:

- Simple, **elegant architecture** that performs well in many image classification tasks.
- **Transfer learning:** Pretrained models are widely available and can be fine-tuned for specific tasks.
- Known for its ability to **capture spatial hierarchies** in images due to deep layers and small receptive fields.

The confusion matrix obtained for the test dataset from the VGG16 Image model is given below:

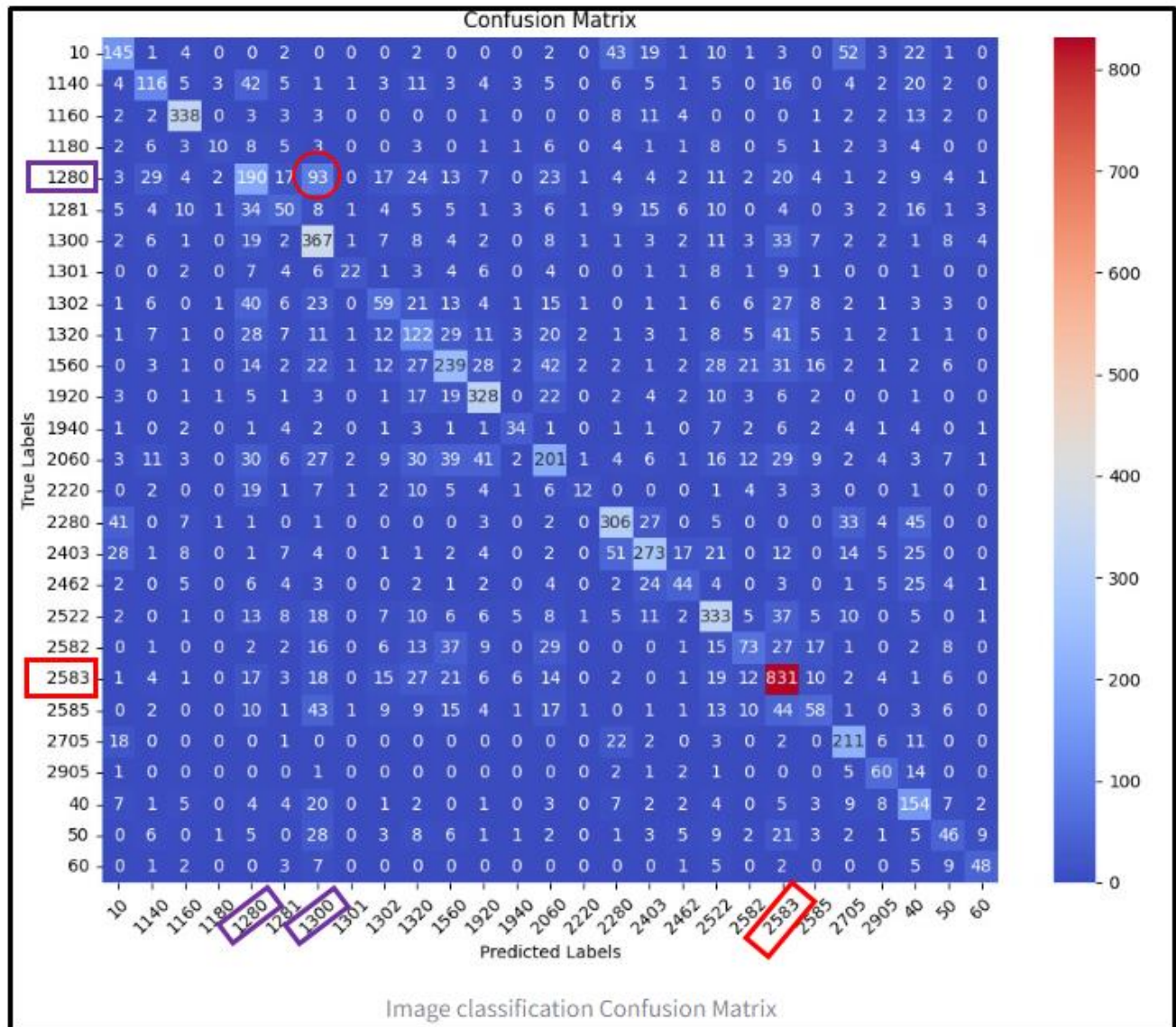


Figure 18. Confusion matrix for image classification model

Classification Statistics

The highest number of correctly classified products was observed in prdtypcode 2583 ("Pool and Accessories"). This category represents the majority class, with a total of 10,209 records in the dataset.

Misclassification Statistics

The class with the highest number of mismatches is Class 1280 (Children's Toys), with a total of 606 mismatches. Notably, most of these mismatches are due to the model incorrectly classifying 93 images as belonging to Class 1300 (Modeling), which suggests that the model has difficulty distinguishing between children's toys and models (figurines). Despite Class 1280 being a majority class, this indicates that there might be significant similarities in features between the two classes, such as shape, color, or context within the images. This observation can guide further analysis of the visual characteristics of both categories to address the issue.

On the other hand, Class 60 (Game Consoles), which is a minority class with only 832 instances, has the lowest number of mismatches at 58 mismatches. This indicates that the model is highly effective at recognizing game consoles, despite their minority status in the dataset. This could be due to distinctive features that help the model easily differentiate game consoles from other categories.

9. FUSION MODEL

Multimodal Model Integration

To further improve classification accuracy, the text and image classifiers were integrated into a unified multimodal system. The late fusion technique was employed to combine the predictions of both models, leveraging the complementary strengths of text and image data. This approach led to better overall performance and contributed significantly to the success of the project.

Feature Fusion

Feature fusion is a technique where multiple feature sets from different models or sources are combined to improve model performance. In the context of machine learning, feature fusion can be used to integrate different types of features, such as those derived from text, image, or other modalities, to enhance the predictive power of the classifier.

Feature Fusion using Voting Classifier

In this approach, multimodal fusion is achieved at the decision level by combining the predictions generated independently by the text-based and image-based models.

The following steps describe the voting-based feature fusion methodology employed in this work:

Step 1: Text-Based Classification using DistilBERT

The textual information of each product, including the title and description, is pre-processed through tokenization and attention masking. The processed text is passed to the **DistilBERT-base-uncased** model, which generates contextual embeddings. These embeddings are fed into a classification layer to produce class probability scores or predicted labels for each product.

Step 2: Image-Based Classification using VGG16

Product images are resized and normalized according to the input requirements of **VGG16**. The images are passed through the VGG16 network pre-trained on ImageNet, with the final classification layers replaced by task-specific dense layers. The model outputs class probabilities or predicted labels based solely on visual features.

Step 3: Independent Model Predictions

Both DistilBERT and VGG16 generate independent predictions for each product instance. Each model acts as an individual classifier, providing its own decision based on the corresponding modality.

Step 4: Voting-Based Fusion

The final product category is determined using a **Voting Classifier** mechanism. The predictions from the text-based and image-based models are combined using either:

- **Hard Voting**, where the class label predicted by the majority of models is selected.
- **Soft Voting**, where the class probabilities from each model are averaged and the class with the highest combined probability is chosen.

Step 5: Final Prediction

The output of the voting classifier represents the final multimodal classification result. This decision-level fusion allows the system to leverage complementary strengths of both textual and visual models.

Step 6: Evaluation and Analysis

The performance of the voting-based fusion model is evaluated using metrics such as accuracy, F1-score, and confusion matrix. The results are compared with unimodal models to assess the effectiveness of the decision-level fusion strategy.

Model Selection for Multimodal Fusion

From both text and image classification models, the models with highest F1-score were chosen for building a fusion model. Therefore, from text classification models, DistilBERT-base-uncased model with a learning rate of $4e-5$ is selected. Likewise, for image classification VGG16 with learning rate of $1e-6$ is chosen for the fusion.

Since a higher F1-score is obtained on the text classification model (0.8343) than image classification model (0.5374), normalized weights will be added to the later fusion model to have better accuracy as text model performs relatively better. The technique used for the fusion of both text classification and image classification model is late fusion with voting classifier using soft voting technique.

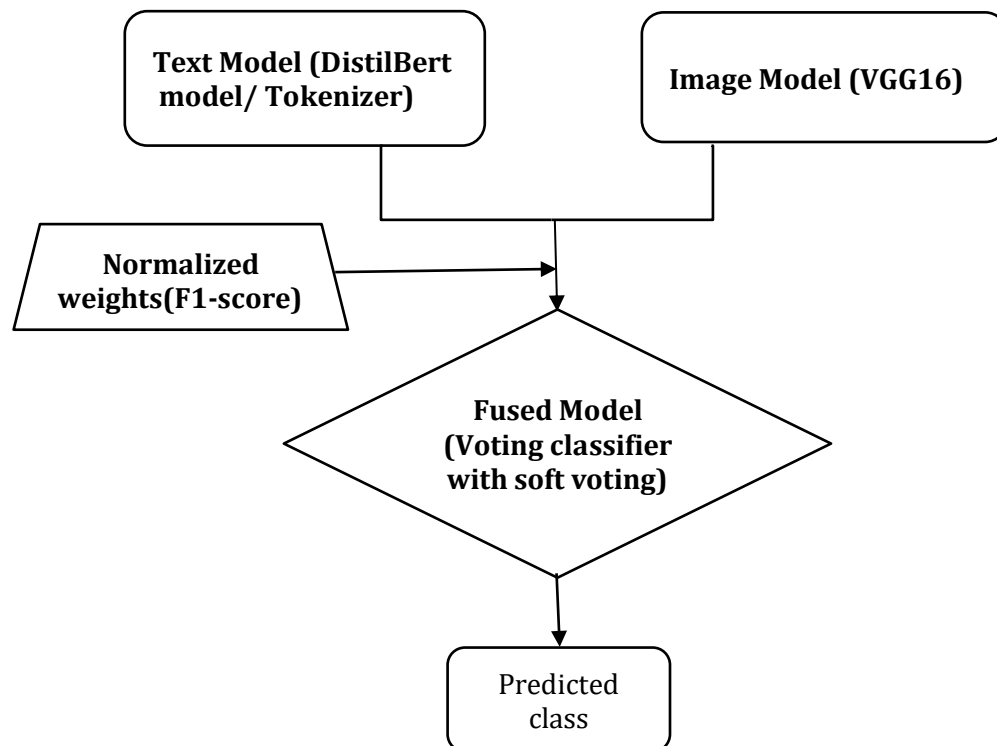


Figure 19. Multimodal development with Late fusion and normalized weights

```
text_input = "Olivia"
img_path = "/content/drive/My Drive/MTech_Dissertation/Models/fused_model/images/image_train/image_1263597046_product_3804725264.jpg"

final_prediction = late_fusion_weighted_soft_voting(
    text_input,
    img_path,
    text_weight=0.61,
    image_weight=0.39
)

print("Final Predicted Class:", final_prediction)
```

... 1/1 0s 60ms/step
Final Predicted Class: 10

Figure 20. Implementation output giving predicted class

As in the above code screenshot, given the “text_input” which is used for text classification and “img_path” for image classification, these are input for “late_fusion_weighted_soft_voting” method. The result is predicted class which is category (Product_type_code).

10. RESULTS

The text classification model (DistilBERT-base-uncased) exhibited an improvement in classification accuracy, particularly for minority classes (those comprising less than 10% of the majority class). This includes categories such as prdtypecode 60 (Game Consoles), 2220 (Pet Store), 1940 (Confectionery), 1301 (Baby Socks, Small Photos), 1180 (Trading Cards), and 2905 (PC Video Games). This suggests that the model has enhanced its ability to correctly classify less-represented categories.

In contrast, despite incorporating image augmentation for minor classes during pre-processing, the image classification model did not show significant improvements for these minority classes. Some of the minor classes remained misclassified, indicating that the current augmentation strategy was not effective for all underrepresented categories.

11. APPLICATION AND FUTURE POTENTIAL OF THE MODEL

The multimodal classification system developed in this project has broad applications, particularly in areas such as product categorization for e-commerce platforms and image-based content tagging. By effectively integrating text and image data, the system can offer more

accurate product classifications, benefit businesses and enhance the user experience.

12. CHALLENGES ENCOUNTERED DURING THE PROJECT

Training the models took longer than anticipated due to our hardware computational limitations and huge datasets, which resulted in more computational time to test the model.

13. FUTURE SCOPE OF THE MODELS

For text classification, employing advanced models combined with varied pre-processing techniques can significantly enhance prediction accuracy.

In the case of the image classification model, most misclassifications arise due to the presence of similar products in multiple classes. Specifically, some classes are subcategories of broader categories, leading to overlap and confusion. To improve prediction accuracy, these subcategories should be merged under a single general category.

For instance, classes such as 10 (Used Books), 2280 (Used Newspapers and Magazines), 2403 (Books, Comics, and Magazines), and 2705 (New Books) all include images of books, making it challenging for the model to correctly distinguish between them.

Enhancing the performance of the image classification model can involve additional pre-processing steps, such as reducing image dimensions, increasing image augmentation, applying image cropping, and utilizing models with deeper CNN architectures, which are better equipped to achieve higher accuracy.

DECLARATION

I hereby declare that the project report entitled “**MULTIMODAL PRODUCT CLASSIFICATION FOR E-COMMERCE CATALOG**” submitted in partial fulfilment of the requirements for the award of the degree of M.Tech in Artificial Intelligence and Machine Learning at Birla Institute of Technology and Science, Pilani, is my original work and has not been submitted previously for any degree or diploma.

All sources of information used in this project have been duly acknowledged.

This work was carried out under the guidance of my project supervisor at **Continental Alloys Middle East Fze** and faculty mentor from BITS Pilani.

Place: Dubai

Date: 28-01-2026



Signature of the Student

Name: FEENAZ TASNEEM SAYED

ID No.: 2023AA05354

APPENDICES – A

Multimodal Development with Late Fusion and Normalized Weights

The proposed multimodal classification framework based on late fusion with weighted soft voting for e-commerce product categorization. The architecture integrates two independent deep learning models that process heterogeneous data modalities text and images before combining their predictions at the decision level.

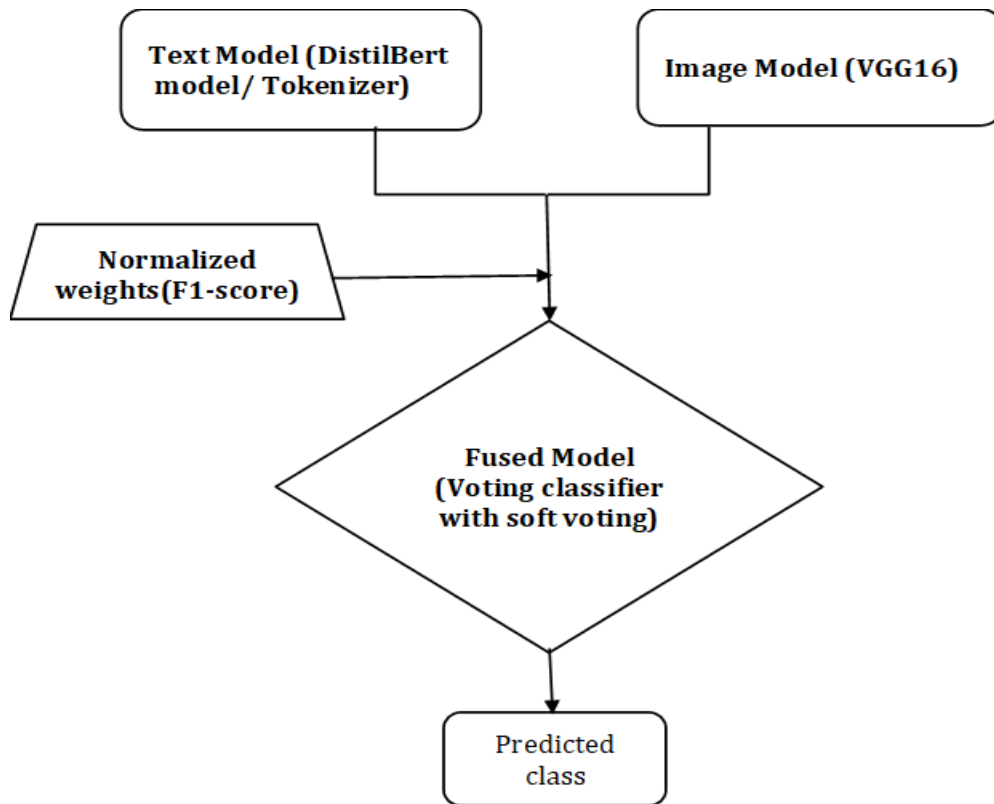


Figure A.1 Multimodal development with Late fusion and normalized weights

The textual modality is handled by a DistilBERT-based text classification model, which processes product titles and descriptions using a pretrained tokenizer and transformer encoder to generate probability scores over product categories. In parallel, the visual modality is processed using a VGG16 convolutional neural network, which extracts discriminative visual features from product images and produces corresponding class probability distributions.

To balance the contribution of each modality, normalized weights derived from validation F1-scores are applied to the probability outputs of both models. These weights reflect the relative reliability of the text and image classifiers and ensure that the more accurate modality has a greater influence on the final decision.

The weighted probability vectors are then combined using a soft voting-based fusion mechanism, where class-wise probabilities from both models are aggregated. The fused model selects the category with the highest combined probability as the final predicted class. This late fusion strategy enables robust and flexible integration of multimodal information while preserving the strengths of individual models, leading to improved classification accuracy in real-world e-commerce environments.

APPENDICES – B

Descriptions of Code Files

1. Exploratory_analysis.ipynb:

Purpose: Conducts exploratory data analysis (EDA) on a combined text and image dataset.

Source Files Used:

- **X_train_update.csv:** Training data for features.
- **Y_train_CVw08PX.csv:** Labels corresponding to training data.
- **X_test_update.csv:** Test dataset for features.
- **translated_text.csv:** File containing preprocessed or translated text data.
- **Train_merged.csv:** A combined dataset with features and labels.
- **df_image_data_20241024.csv:** Metadata or extracted features from image data.
- **Additional Inputs:** Image data taken from online resources.

2. DistilBERT_base_uncased_model.ipynb:

Purpose: Implements a text classification model using the DistilBERT base uncased pre trained model.

Source Files Used:

- **BERT_dataset.csv:** Dataset prepared for training and testing the text classifier.
- **Output:** Trained DistilBERT model, tokenizer, and evaluation metrics.

3. Vgg16_image_model.ipynb:

Purpose: Implements an image classification model using the VGG16 convolutional neural network architecture.

Source Files Used:

- **image_train folder (within the image data folder):** Contains training images for the model.
- **Output:** Trained VGG16 model and evaluation metrics.

4. late_fusion_model.ipynb:

Purpose: Combines predictions from the DistilBERT text model and the VGG16 image model using a voting classifier (weighted and soft voting).

Source Files Used:

- Saved DistilBERT and VGG16 models.
- Tokenizer and label encoder used during text and image model training.
- Output: Final classification results, combining text and image modalities.

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GLOSSARY

AI – Artificial Intelligence (pp. 2, 6)

A branch of computer science that enables machines to perform tasks requiring human intelligence, such as learning, reasoning, perception, and decision-making.

CNN – Convolutional Neural Network (pp. 5, 7, 32, 33, 42)

A deep learning architecture widely used for image processing and visual feature extraction in product classification systems.

Transformer (pp. 7, 9, 44)

A neural network architecture based on attention mechanisms, commonly used for processing sequential data such as text.

BERT – Bidirectional Encoder Representations from Transformers (pp. viii, x, 1, 27, 31, 34, 39, 46, 47, 50)

A transformer-based language model used for extracting contextual textual features from product descriptions.

DistilBERT (pp. X, 1, 31, 34, 34, 50)

A lightweight and computationally efficient version of BERT that retains most of its language understanding capability.

Feature Fusion (pp. 2, 3, 6, 38, 39)

The process of combining features extracted from multiple modalities to improve classification performance.

Late Fusion (pp. ix, 38, 40, 44, 45)

A fusion technique where predictions from separate models are combined at the decision level.

EDA – Exploratory Data Analysis (pp. 8, 50)

An approach used to analyze datasets through statistics and visualizations to understand patterns

and data distribution.

Class Imbalance (pp. 2, 5, 7, 27, 33, 51)

A condition where some product categories have significantly more samples than others, affecting model learning.

F1-Score (pp. viii, 4, 9, 28, 29, 32, 40, 51)

An evaluation metric that balances precision and recall, especially useful for imbalanced classification problems.

Confusion Matrix (pp. 9, 35, 37, 51)

A table used to evaluate classification performance by comparing predicted and actual class labels.

Transfer Learning (pp. 27, 33, 51)

A technique where pre-trained models are reused for a related task to improve performance and reduce training time.

Checklist of Items for the Final Dissertation / Project / Project Work Report

This checklist is to be attached as the last page of the final report.

This checklist is to be duly completed, verified and signed by the student.

1.	Is the final report neatly formatted with all the elements required for a technical Report?	Yes
2.	Is the Cover page in proper format as given in Annexure A?	Yes
3.	Is the Title page (Inner cover page) in proper format?	Yes
4.	(a) Is the Certificate from the Supervisor in proper format? (b) Has it been signed by the Supervisor?	Yes Yes
5.	Is the Abstract included in the report properly written within one page? Have the technical keywords been specified properly?	Yes Yes
6.	Is the title of your report appropriate? The title should be adequately descriptive, precise and must reflect scope of the actual work done. Uncommon abbreviations / Acronyms should not be used in the title	Yes
7.	Have you included the List of abbreviations / Acronyms?	Yes
8.	Does the Report contain a summary of the literature survey?	Yes
9.	Does the Table of Contents include page numbers? (i). Are the Pages numbered properly? (Ch. 1 should start on Page # 1) (ii). Are the Figures numbered properly? (Figure Numbers and Figure Titles should be at the bottom of the figures) (iii). Are the Tables numbered properly? (Table Numbers and Table Titles should be at the top of the tables) (iv). Are the Captions for the Figures and Tables proper? (v). Are the Appendices numbered properly? Are their titles appropriate	Yes Yes Yes Yes Yes Yes Yes
10.	Is the conclusion of the Report based on discussion of the work?	Yes
11.	Are References or Bibliography given at the end of the Report? Have the References been cited properly inside the text of the Report? Are all the references cited in the body of the report	Yes Yes Yes
12.	Is the report format and content according to the guidelines? The report should not be a mere printout of a PowerPoint Presentation, or a user manual. Source code of software need not be included in the report.	Yes

Declaration by Student:

I certify that I have properly verified all the items in this checklist and ensure that the report is in proper format as specified in the course handout.

Place: Dubai

Signature of the Student

Date: 28-01-2026

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